Runs and scripts

$E = mc^2$

$x_i^2 + y_j$

$^{14}_{6}\mathrm{C} \rightarrow {}^{14}_{7}\mathrm{N}$

$\alpha + \beta = \gamma$

$\sin\theta + \cos\theta$

$area = \pi r^2$

Fractions and radicals

$$y = \frac{a}{b}$$

$$\frac{\frac{1}{x}}{2} + 1$$

$$\binom{n}{k}$$

$$\sqrt{x+1}$$

$$\sqrt[3]{8} = 2$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Delimiters and limits

$$s = \frac{n\kappa n + 1\lambda}{2}$$

$$|x - 1| \leq 2$$

$$\sin^2 x + (a + b)^2$$

$$\left[\frac{a}{b}\right]$$

$$\{x|y\}$$

$$\lim_{n \to \infty} \frac{1}{n}$$

Matrices, bars, and boxes

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$a_{n+1} = \sqrt{a_n + 1}$$

$$\underline{z} + \underline{w}$$

$$\underbrace{a + b}_{} + \overset{def}{x}$$

$$a = b + \boxed{c}$$

$$x + y = 1$$
$$x - y = 0$$